

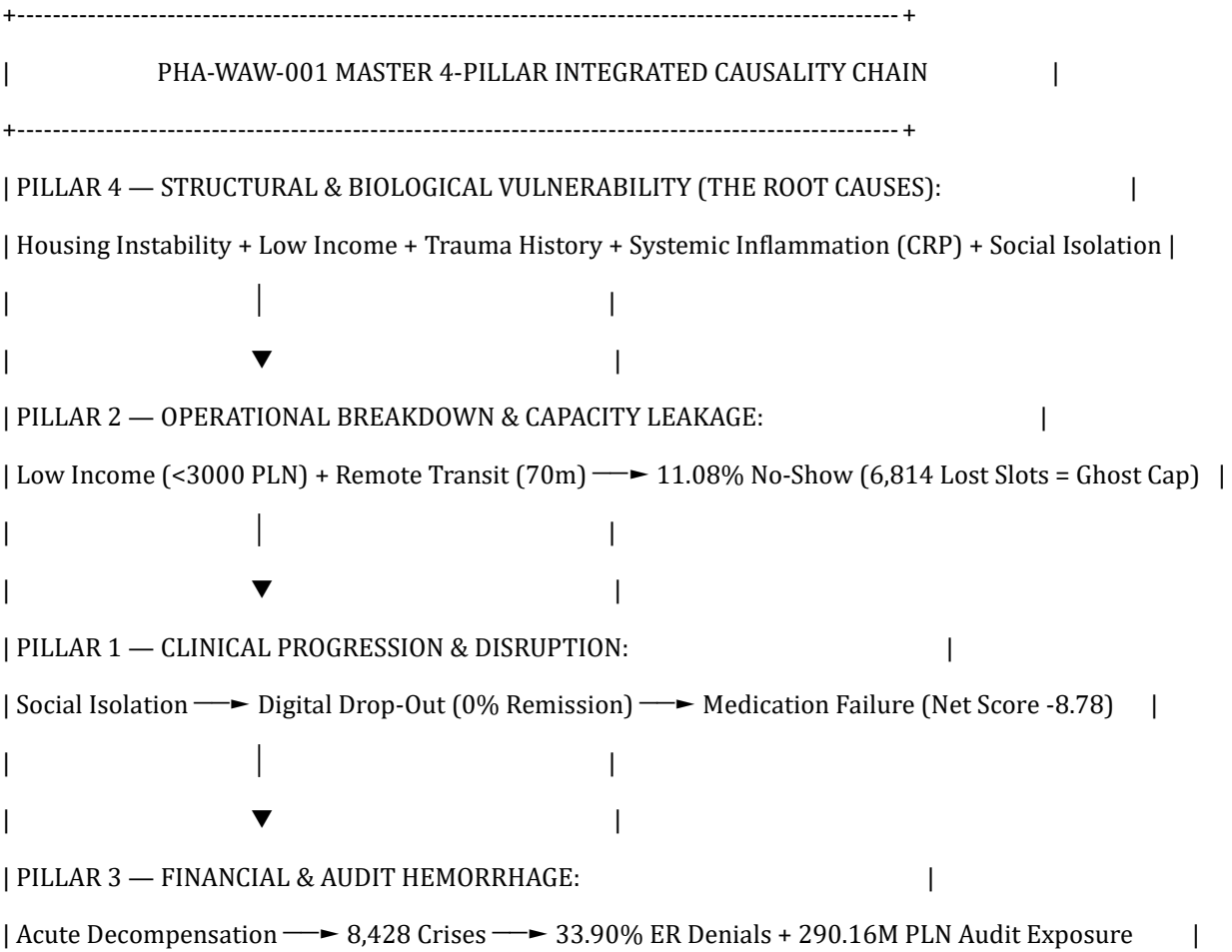
Social Determinants of Health (SDOH), Biomarkers & Structural Vulnerability

TARGET PILLAR: PILLAR 4 — Social Determinants of Health, Biomarkers & Structural Vulnerability

Architect for the **PHA-WAW-001 Program**, Pillar 4 has been constructed and expanded to achieve maximum analytical depth and integrative linkage across all **17 verified physical schema tables** (SDOH, Social_Graph, Biomarkers, Clinical_Notes, Clinical_Assessments, Patients, Crisis_Events, Financials, Treatments, Longitudinal_Sessions, Outcomes).

In this expanded blueprint, Pillar 4 serves as the **"Root-Cause Engine"** that resolves the core empirical anomalies identified across the preceding three pillars:

- **The mystery of the 25,969 patients** who failed in digital engagement and recorded **0% remission** (Pillars 1 & 3).
- **The mystery of the peak non-attendance rate (14.39%)** and unutilized capacity loss in remote catchment areas (Pillar 2).
- **The mystery of the concentration of 8,428 acute crises, 4,228 hospitalizations**, and the elevated **33.90% ER claim rejection rate** (Pillars 1 & 3).
- **The mystery of medication alteration failures** and the observed **-8.78 drop in net clinical improvement** (Pillars 1 & 2).



THE 4 EXPANDED STRATEGIC INQUIRY DOMAINS (PILLAR 4)

1. MODULE 4.1 (ANA-401): Housing Instability, Low Income & Transit Friction as Structural Drivers of Capacity Loss and Financial Leakage

- **Verified Physical Schema Anchors:**

- dbo.SDOH (TBL-015: Housing_Instability, Education_Level)
- dbo.Patients (TBL-012: Monthly_Income_PLN, Distance_to_Clinic_km, Transport_Time_mins, District)
- dbo.Appointments (TBL-001: Status, Wait_Time_Days)
- dbo.Financials (TBL-007: Total_Billed_PLN, Claim_Rejected_Flag)

- **Analytical Hypothesis & Causality (The "Why"):**

Patients experiencing housing instability (Housing_Instability = 1) and severe financial constraints (Monthly_Income_PLN < 3000 PLN) in remote catchment zones (≥ 15 km and transit time > 70 mins) incur disproportionate economic and logistical opportunity costs. When basic survival needs conflict with clinic attendance, appointments are missed. This explains why the no-show rate escalated to **14.39%** for remote prompt appointments in Pillar 2, generating **114.54M PLN** in unutilized capacity leakage.

- **Expected Impact & Strategic Outputs:**

Isolates high-vulnerability catchment districts to deploy transit-subsidized care pathways or route high-friction cohorts into **Subsidized Telehealth**, reclaiming up to **6,814 unutilized appointment slots**.

2. MODULE 4.2 (ANA-402): Social Capital, Family & Peer Support Networks as a Buffer Against Digital Engagement Failure

- **Verified Physical Schema Anchors:**

- dbo.Social_Graph (TBL-016: Family_Support_Score, Caregiver_Involved_Flag, Peer_Support_Group_Flag)
- dbo.SDOH (TBL-015: Social_Isolation_Score)
- dbo.Longitudinal_Sessions (TBL-009: App_Usage_Hours, Homework_Completed, Telehealth_Flag)
- dbo.Treatments (TBL-017)
- dbo.Outcomes (TBL-011: Trajectory_Outcome)

- **Analytical Hypothesis & Causality (The "Why"):**

In Pillars 1 and 3, 25,969 patients in the passive engagement tier achieved 0% full remission, consuming 88.76M PLN with zero recovery return. Severe social isolation (Social_Isolation_Score) and the absence of a designated caregiver (Caregiver_Involved_Flag = 0) are the underlying drivers of digital non-compliance. Conversely, active peer support groups (Peer_Support_Group_Flag = 1) and family scaffolding maintain digital engagement above 8.5 hours/week, unlocking the **31.08% full remission rate** at an optimized cost of **11,134 PLN per remitted patient**.

- **Expected Impact & Strategic Outputs:**

Informs clinical pathways to mandate digital caregiver companions and peer group integration prior to enrolling isolated patients into standalone digital therapeutics.

3. MODULE 4.3 (ANA-403): Neurobiological Stress (CRP/Sleep), Prior Trauma History & Pharmacological Treatment Resistance

- **Verified Physical Schema Anchors:**

- dbo.Biomarkers (TBL-002: CRP_Level, Sleep_Hours_Avg, Genetic_Risk_Score, Wearable_Stress_Index)
- dbo.SDOH (TBL-015: Trauma_History_Flag)
- dbo.Treatments (TBL-017: Medication_Change_Flag, Side_Effect_Severity, Therapy_Type)
- dbo.Outcomes (TBL-011: Trajectory_Outcome)
- dbo.Financials (TBL-007: Total_Billed_PLN)

- **Analytical Hypothesis & Causality (The "Why"):**

Pillar 1 established that iterative drug changes coupled with severe side effects resulted in a **-8.78** net clinical score decline. Patients with documented trauma history (Trauma_History_Flag = 1) frequently exhibit systemic neuroinflammation (CRP_Level > 3.0 mg/L) and severe sleep disruption (Sleep_Hours_Avg < 5.0 hrs), inducing biological resistance to standard monotherapies. Iterative medication switching without addressing trauma or systemic inflammation inflates billing without improving outcomes.

- **Expected Impact & Strategic Outputs:**

Establishes baseline inflammatory and sleep screening to route high-stress, trauma-exposed patients directly into specialized trauma modalities (such as EMDR/DBT), avoiding redundant pharmaceutical switching cycles.

4. MODULE 4.4 (ANA-404): Immigrant Vulnerability, Clinical Note Sentiment Discordance & Proactive Acute Crisis Prediction

- **Verified Physical Schema Anchors:**

- dbo.SDOH (TBL-015: Immigration_Status, Social_Isolation_Score)
- dbo.Clinical_Notes (TBL-004: Sentiment_Score, Self_Harm_Mentions, Hallucination_Mentions, Substance_Use_Mentions)
- dbo.Clinical_Assessments (TBL-003: Suicide_Risk_Score, Score)
- dbo.Crisis_Events (TBL-006: ER_Visit_Flag, Hospitalization_Flag)
- dbo.Financials (TBL-007: Total_Billed_PLN, Claim_Rejected_Flag)

- **Analytical Hypothesis & Causality (The "Why"):**

In Pillars 1 and 3, 8,428 crisis events and 4,228 hospitalizations were concentrated in severe depression, generating a 33.90% claim denial rate in emergency services. Immigrant populations and socially isolated cohorts underreport distress on structured instruments (Suicide_Risk_Score) due to language barriers or stigma. However, true clinical decompensation manifests early in unstructured clinical notes (Sentiment_Score < -0.40) and self-harm mentions prior to emergency admission.

- **Expected Impact & Strategic Outputs:**

Enables the deployment of an automated real-time NLP surveillance layer on clinical notes for vulnerable populations, intercepting acute decompensations and mitigating the **33.90% ER claim denial penalty**.

 MASTER 4-PILLAR INTEGRATION MATRIX

Strategic Pillar	Analytical Domain	Cross-Pillar Input & Dependency	Strategic Executive Value
Pillar 1: Clinical Outcomes	Treatment Efficacy & Continuity	Net score decay (−8.78) vs. high digital engagement (74.76% positive response).	Links clinical recovery to trauma history, systemic inflammation, and family support (P4).
Pillar 2: Operational Efficiency	Capacity & Latency Bottlenecks	11.08% remote no-show rate, 70.8% clinician overload, 54.26% severe scheduling delay.	Explains non-attendance via income constraints, transit friction, and housing instability (P4).
Pillar 3: Financial Security	Revenue Cycle & Value-Based ROI	280.75M PLN total leakage; high digital engagement ROI (11.1K vs. 173K PLN).	Directs capital investment toward patient cohorts with verified structural engagement scaffolding (P4).
Pillar 4: SDOH & Biomarkers	Structural & Biological Root Causes	Integrates biological, social, and psychological determinants.	Establishes a proactive, preventive healthcare delivery model protecting clinical outcomes and revenue.